

Sentiment-Based Book Recommendations: A Machine Learning and Natural Language Processing Approach to Enhance Reading Experience

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Abstract. Sentiment Analysis (SA) and emotion detection/recognition within text data are dynamic and interconnected fields garnering significant scholarly attention. While SA focuses on distinguishing neutral, positive, or negative tones, emotion analysis delves deeper, aiming to identify specific emotions like happiness, surprise, grief, disgust, fear, and anger. This study proposes a novel approach for book recommendation by combining emotion and genre prediction based on user reviews. The system integrates tweet emotion data and Amazon book reviews, processes text using NLP techniques, applies TF-IDF for feature extraction, addresses class imbalance via ADASYN, and uses Logistic Regression (LR), Support Vector Machine (SVM), and Random Forest (RF) classifiers evaluated using stratified k-fold validation. Google Book API enables genre prediction. SVM achieved 96.92% accuracy, showing significant performance gain. This demonstrates the potential of combining sentiment and genre for smarter book recommendations.

Keywords: Sentiment lexicon · Emotion detection · Book recommender system · Machine learning · Natural language processing

1 Introduction

Sentimental analysis is a common machine-learning method that aids in recognizing emotions. It enables businesses to get vital information about their consumers' preferences via numerous social media platforms, polls, and website assessments [1]. The study of assessing client attitudes, expressions, preferences, and dislikes toward diverse things like products, services, companies, and people is driven by sentiment analysis. People can share their opinions and experiences with numerous products via social media and e-commerce platforms such as Amazon, Flipkart, and others [2]. Suppose someone posted a recipe for Italian pasta on a recipe website and noted, "Though I attempted the same method as stated, the meal tasted excessively salty." We can apply sentiment analysis to determine whether or not the reviewer liked the dish". Too salty" is a negative feature that indicates the client did not desire the dish [3].

Making decisions can be tricky, and often, we turn to others for advice. This is especially true when it comes to choosing a product, book, or restaurant [4]. Online reviews have become an invaluable tool in this process, offering us insights into other people's experiences and satisfaction levels [5]. But how can we analyze and understand these vast amounts of opinions? That's where sentiment analysis comes in. This powerful technique uses artificial intelligence to automatically analyze text and identify the sentiment behind it – positive, negative, or neutral and so on [6]. Traditionally used to understand customer feedback and social media trends, sentiment analysis is now being integrated with recommender systems [7]. By incorporating sentiment analysis, these systems can go beyond simply analyzing ratings and delve deeper into user opinions, leading to more personalized and relevant recommendations [8]. While sentiment analysis has shown promise in various areas, its application in book recommendation systems remains relatively unexplored. This presents a unique opportunity to improve the accuracy and effectiveness of book recommendations, ultimately helping readers discover their next great read [9].

A book recommender system is designed to suggest books to readers based on their past interests, offering recommendations similar to their previously enjoyed titles [10]. These systems are commonly employed by platforms that offer e-books, including Google Play Books, Goodreads, and Open Library [11]. A book recommender system, designed to suggest books based on past interests, typically recommends titles similar to those previously enjoyed [12]. These systems are commonly used by platforms like Google Play Books, Goodreads, and Open Library, offering e-books [13]. DistilBert is a compact version of BERT created through knowledge distillation during pre-training, making it 40% smaller, 60%

faster and still retaining about 97% of BERT’s performance. Moreover the result is a lighter, cheaper and efficient model suitable for NLP tasks under limited computational resources [45]. Recent research emphasizes how pretrained language models, such as BERTweet, can be employed to leverage social media data for public health surveillance. For instance, BERTweet has been utilized to classify tweets related to vaping, capture commercial versus non-commercial content, and assess sentiment with relevant accuracy, yielding AUROCs of 99.3% and weighted F1 scores over 97%. Overall, BERTweet was found to be accurate and reliable in all classification conditions, outperforming traditional models such as LSTM. These results point to the potential for pretrained transformers to be utilized to conduct sentiment analysis, especially on user-generated text where there is noise in the data such as tweets [46]. Proposed zero-shot cross-lingual NER proceeds with parallel corpora with XLM-RoBERTa and an entity alignment model; pseudo-labeled target-language data are generated, but not based on translation. It reaches competitive F1 scores on 4 languages, with a noisy classification loss function adjusted to process noisy labeling, and points out the effects corpus size and domain can have on transfer learning [47]. This research proposes enriching this analysis by recommending books tailored to readers’ emotions and preferred genres. Combining psychology and the lexical database ‘WordNet’, the system recognizes emotions in user reviews, predicting happiness, sadness, fear, anger, and surprise. These emotions inform recommendations aligned with individual preferences. Several preprocessing steps are implemented, including creating a new emotion class, cleaning text, balancing class distribution, and evaluating multiple machine learning classifiers for their effectiveness in emotion prediction from crowdsourced ‘Twitter data’. Google’s Book API is then integrated to allow genre forecasting. Careful tuning optimizes performance, showcasing the potential of sentiment analysis and genre prediction innovations to transform book recommendations. This customized approach employs machine learning for emotion detection and assesses its effectiveness, aiming to introduce an emotionally intelligent system. For example, if a user’s mystery book reviews are very positive while their romance reviews are more negative, the system might recommend a new mystery over a romance novel. This personalized approach serves up books that someone is likely to connect with emotionally. This research makes several significant contributions. The key contributions of this study are:

- Integrating two datasets by correlating emotion data and book review data from Twitter and Amazon.
- Implementing advanced preprocessing techniques, including tokenization, lemmatization, and TF-IDF.
- Addressing model bias through emotion class reduction and ADASYN-based class balancing.
- Performing a comparative evaluation of three machine learning classifiers (LR, RF, SVM) using stratified k-fold cross-validation.

The preference for using classical machine learning models like Support Vector Machine, Random Forest, and Logistic Regression is driven by practical and theoretical reasons. These models are more interpretable than intricate deep learning architectures, enabling a better understanding and explanation of decision-making processes—a consideration that is crucial for establishing trust and transparency in recommendation systems. Furthermore, classical models require much less computational power, making them easier to deploy in hardware-constrained environments or where real-time processing is required. Together with effective strategies like data balancing using ADASYN and stringent evaluation via stratified K-Fold cross-validation, these models show strong and stable performance on imbalanced datasets, as seen in the high accuracy obtained in emotion prediction. This trade-off of efficiency, interpretability, and predictive capability warrants the methodological preference for classical machine learning models in this research.

2 Literature Review

Before entering into the complexities of our research, we must contextualize our conversation by reviewing important scholarly publications and studies. This introductory phase will provide a core understanding of what we have found so far from previous research regarding this topic. The overall overview of some papers is given below:

This research approach is rooted in formal psychological and affective theory, increasing the interpretability and relevance of the recommendation system over pure data-driven approaches. By drawing on Affective Computing concepts, the system is set to sense and react to users’ emotional states from textual reviews to provide recommendations with emotional resonance. By drawing on Self-Determination

Theory (SDT), there is a theoretical rationale for connecting emotional experience with intrinsic motivation and reading behavior, such that emotionally personalized recommendations can better serve users' autonomy and engagement. Such theoretical integration not only theoretically grounds the system but also provides opportunities to develop more human-centered, emotionally intelligent recommendation systems that couple technological capability with users' psychological needs.

The study of Wayesa et al. [12] contributed to this field by proposing a pattern-based hybrid recommendation approach that uses semantic relationships to capture the relationships as knowledge-based book recommendations to readers in a digital library. The system employs Content-based Filtering (CBF) and Collaborative Filtering (CF) with semantic relationships to capture the similarity of the books that the new user had retrieved. The patterns were grouped in a semantically equivalent manner using the Clustering method. The effectiveness of the suggested models is examined through a series of extensive tests employing Information Retrieval (IR) evaluation criteria. The paper uses the good books dataset, which is a commonly used dataset in the domain of recommender systems. The dataset contains the results of real users' interactions with the recommender system. It can recommend books using the user profile. The results showed that clustering-based recommendation had higher accuracy, with a precision of 76.4%, recall of 37.4%, and F1-score of 56.9%, compared to individual user recommendation. However, the paper does not discuss the scalability of the proposed model. The proposed model is evaluated on a relatively small dataset, and it is unclear how it would perform on a larger dataset. Lin et al. [13] research constitutes a significant endeavor in the domain of text classification, particularly focusing on the representation and classification of text utilizing a combination of classifiers and pre-trained models. They talk about the difficulties in text representation and offer a framework for text classification tasks that makes use of pre-trained models, particularly BERT. The authors emphasize how BERT, which is trained using next sentence prediction mechanisms and masked language modeling, is effective in producing satisfactory results. They collected dataset from Instagram, Twitter, and travel websites. The proposed model in the study achieved an accuracy metric score of 85.17, which was the highest among all the models evaluated. Some key limitations or challenges of this paper included: Small, imbalanced Indonesian datasets of short text; limited model evaluation across languages, dataset sizes/types, and classification complexities. Model complexity increases computation time. Lacks optimization for practical usage and human evaluation to confirm performance gains.

The study conducted by Addanki et al. [14] contributes significantly to the field by incorporating sentiment analysis into a book recommender system. The work involves using sentiment analysis to translate textual reviews into numerical ratings. It explores the role of contextual information in sentiment-based recommendation systems and discusses the use of hybrid recommendation approaches for e-commerce platforms and movie recommendation models. Sentiment analysis is pivotal in enhancing the accuracy of these recommendation frameworks. The researchers used two well-known datasets for their study. These included Amazon reviews of expensive meals and movies, and the results were reported to be very satisfying. This study mentions the use of several algorithms, including collaborative filtering, the Prefix Span sequential pattern mining technique, the Topseq rule mining technique, K Means, KNN (k nearest neighbor), proposed DBSCAN, and OPTICS Algorithm. Additionally, the researchers used VSM and TF IDF to analyze thoughts and develop recommendation systems based on sentiment analysis. In this study, OPTICS algorithm performed best and the highest accuracy was 91%. In summary, key limitations include small or narrow datasets, lack of real user evaluation, limited comparative analysis, and a need to evaluate robustness over time and changing data. Addressing these could strengthen the evidence for the usefulness of the proposed techniques. The research conducted by Karabila et al. [15] aims to enhance Collaborative Filtering-based Recommender Systems through the integration of deep learning-based sentiment analysis applied to user reviews. It employs BiLSTM networks for contextual sentiment modeling. The BiLSTM sentiment model is combined with user and item-based collaborative filtering approaches using ensemble learning. Evaluations on Amazon product review datasets show significant accuracy improvements compared to just collaborative filtering or using SVM for sentiment analysis. They got the accuracy around 91%. The key novelty is leveraging modern deep learning sentiment techniques to improve traditional recommender systems. Limitations include testing on only books and music datasets and mainly comparing to SVMs. But the work clearly establishes the value of state-of-the-art sentiment analysis for boosting recommender accuracy.

In their study, Dogra et al. [16] looked at sentiment classification in a banking, government and international news context in India. They compared a deep contextual model (DistilBERT) to a traditional context-independent model (TF-IDF) for multiclass sentiment classification (positive, negative, neutral).

DistilBERT was fine-tuned and input to four supervised machine learning classifiers namely: Random Forest, Decision Tree, Logistic Regression, Linear SVC, and analogous experiments were performed with TF-IDF as benchmark. Their experiments found that DistilBERT significantly outperformed TF-IDF because it was able to transfer semantic knowledge to the financial news domain. Random Forest using DistilBERT was the best performing classifier with overall accuracy of 78%, which was 7% higher than TF-IDF with Random Forest. This analysis shows how effective combining contextual embeddings and ensemble-based classifiers for sentiment analysis in specialized domains such as financial and government news.

Nguyen et al. [17] proposed BERTweet, the first large-scale language model that is pretrained on 850M English Tweets. BERTweet is pretrained using the technique of RoBERTa, and the pretrained model was tokenized using a Byte-Pair Encoding tokenizer to sample and identify the unique linguistic features of Twitter, including abbreviations, hashtags, and unconventional use of syntax. When evaluated on various social media data text classification tasks, including sentiment classification, part-of-speech tagging, and named entity recognition (NER), BERTweet demonstrated state-of-the-art results that outperformed general purpose models like BERT and RoBERTa on Twitter datasets. Further, Nguyen et al. asserted the importance and impact of domain-specific pretraining on noisy, user-generated data. The limitations of BERTweet for commercial use included that it was restricted to English Tweets and therefore, offered lower user-appeal for multilingual. Fan et al. [18] formulated the Multiparty Dialog Discourse Parsing (MDDP) problem, which is about identifying discourse links and relations between utterances. Previous efforts dealt with the problems of data sparsity through multitask learning, however those works heavily used fine-grained annotations, which limits practical feasibility. They proposed a Dynamic Task-Adaptive Graph Transformer with Difficulty-Aware Task Scheduling (DTGT-DTS) to overcome this limitation, and included the related tasks of addressee recognition and speaker identification which relate to MDDP through discourse links and maintained private links in those tasks. The DTGT task shared links as well as links specific to the task, while the DTS task was difficulty aware and could make adaptations to the training preparations when given the task difficulty. Evaluations on the database Molweni (78245 links), and STAC (12691 links), shown significant and positive gain of: +6.07% and +5.31% for link identification, and +7.27% and +6.02% for relation recognition respectively. This illustrates that the DTGT-DTS method showed efficient gains in discourse parsing without additional annotated resources.

XLM-RoBERTa, a multilingual transformer model trained on a vast amount of filtered Common-Crawl data (2.5TB) across 100 languages, was introduced by Conneau et al. in [19]. In contrast to mBERT, the authors used SentencePiece for tokenization and a large-scale masked language modelling objective. XLM-RoBERTa was evaluated against cross-lingual benchmarks in XNLI, MLQA, and NER; it surpassed both mBERT and XLM models, achieving the state-of-the-art results for a variety of multilingual understanding tasks. The authors also demonstrate the zero-shot cross-lingual transfer when the model was evaluated outside cross-lingual transfer learning paradigms. The main limitation is the resource requirements: training a model which required many resources would make copying the study for low-resource research environments a significant practical challenge.

The study by Belogianni [20] focuses on implementing a recommendation system utilizing a case-based approach where specific marks or goals are identified by the user, and items with similar characteristics are retrieved using similarity metrics. The system involves comparing book and customer vectors using similarity measures, modeling books, customers, and transactions in a two-layer graph, and implementing a graph search engine to discover relationships between items. The approach focuses on the creation of a recommender system based on collaborative filtering and sentiment analysis to recommend products to customers based on their previous ratings for other products. Surprise library is used for the creation and evaluation of recommender systems with algorithms such as SVD and KNN being examined. They got highest accuracy 89% by SVD. This recommendation systems face a few limitations, including challenges with new user recommendations due to the lack of past user references and ratings for training. This makes it difficult to generate valuable and accurate recommendations for new customers. In the research conducted by Singh et.al [21] published in Human-centric Computing and Information Sciences, the primary focus lies in optimizing sentiment classification accuracy through the application of various machine learning approaches. Four classifiers are utilized - Naive Bayes, J48, BFTree, and OneR. Experiments leverage three manually annotated datasets, including Amazon product reviews and IMDB movie reviews. Following text preprocessing, unigrams and bigrams are extracted as features. Evaluation

metrics comprise precision, recall, accuracy, and F-measure. Results demonstrate the OneR classifier as most precise, achieving 91.3% precision, 97% F-measure, and 92.34% classification accuracy. Naive Bayes exhibits fastest learning while J48 best balances true and false positives. The small-scale experiments provide initial insights but further evaluation is needed on larger and more diverse datasets to establish broader applicability and generalizability.

All studies reviewed in this context endorse the utilization of machine learning and deep learning models, with the most commonly used data set being Amazon. These studies employ various algorithms, with OPTICS being the most frequently applied. Additionally, some studies leverage sentiment analysis to enhance the quality of their recommendation systems. However, many of these studies face challenges such as small datasets, imbalanced classes, lack of significance testing, and a dearth of comparative studies. Furthermore, some encounter suboptimal performance with their applied algorithms. Table 1 provides a summary of these studies.

Table 1: Summarized literature review of book recommendation systems using machine learning.

Studies	Dataset	Model	Performance	Limitations
[12]	Good books dataset	Pattern-based Word Em-bedding	56%	Scalability issue; small dataset was used
[13]	Instagram, Twitter, travel websites	BERT + Bi-LSTM, TCN	85%	Imbalanced datasets; lack of optimization
[17]	MovieLens-1M, Book-Crossing, Bing-News	ReBKC	72%	Smaller dataset was used
[18]	GoodReads books (Kaggle)	K-means + Cosine Distance	52.84%	No baseline comparisons; poor dataset size/UI
[19]	Amazon.com via web crawlers	Logistic Regression	80%	Limited to 221 books/products
[47]	Movie reviews, product views	re-SVD	82%	Limited feature engineering; small-scale datasets
[48]	Twitter data	Naive Bayes + uni-grams/bigrams	82.7%	No contextual embeddings; limited generalization
[49]	Web Product reviews	Feature-based sentiment analysis	84%	Relatively small dataset; domain-specific features
[50]	Movie reviews	PMI-IR Semantic Orientation	74%	Unsupervised method; sensitive to noisy terms

This paper aims to bridge the gap in the existing literature by proposing a new approach that involves combining two datasets, implementing rigorous data preprocessing, addressing class imbalances, and improving performance through the application of traditional machine learning methods, as opposed to more complex deep learning models and others.

3 Proposed Method

The focus of this study is on presenting a comprehensive book recommendation system using machine learning approach. Data gathering, data pre-processing, predicting emotions using machine learning model with model fine-tuning approach are followed to develop the proposed system. The model culminates in a tailored recommendation system for users, leveraging emotion and genre predictions to offer personalized book suggestions aligned with individual preferences and emotion. The overview of the working process is presented in Figure 1.

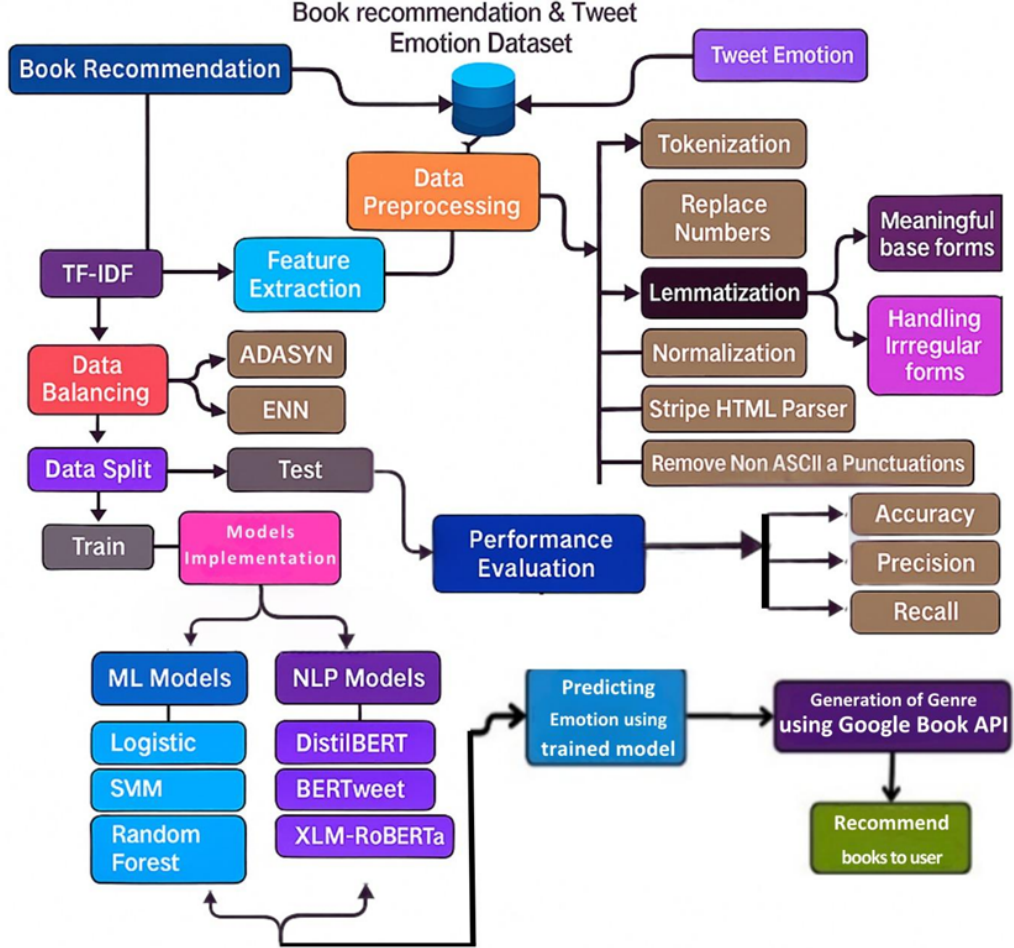


Fig. 1: The workflow diagram of the book recommendation system

3.1 Dataset and Analysis

A. Tweet Emotion Dataset The emotion dataset comprises 39,716 tweet IDs along with corresponding sentiment and content [22]. It encompasses 13 labeled emotions. Figure ??(a) illustrates the distribution of all the original emotion labels in the tweet emotion data. Notably, among the 14 emotions, *neutral* and *worry* have the highest occurrences, while *anger* is the least represented.

In this study, we restricted our work to four classes, as the remaining classes did not contain sufficient data for reliable training. Data augmentation techniques were considered; however, due to the limited availability and diversity of samples in those classes, augmentation could not effectively address the scarcity issue. Furthermore, the dataset exhibited class imbalance, with certain classes being significantly

underrepresented compared to others. These constraints influenced our experimental design and analysis, ensuring that our focus remained on the classes with adequate data to achieve meaningful results. The four considered classes are - neutral, happiness, sadness, worry. The class distribution is displayed in Figure 2 (b). The Figure 2 (b) indicate the dataset is skewed or imbalance of classes. To resolve this imbalance dataset is explain (using ADASYN and ENN) in the data preprocessing section. The sample data of Tweet emotion dataset is tabulate in Table 2.

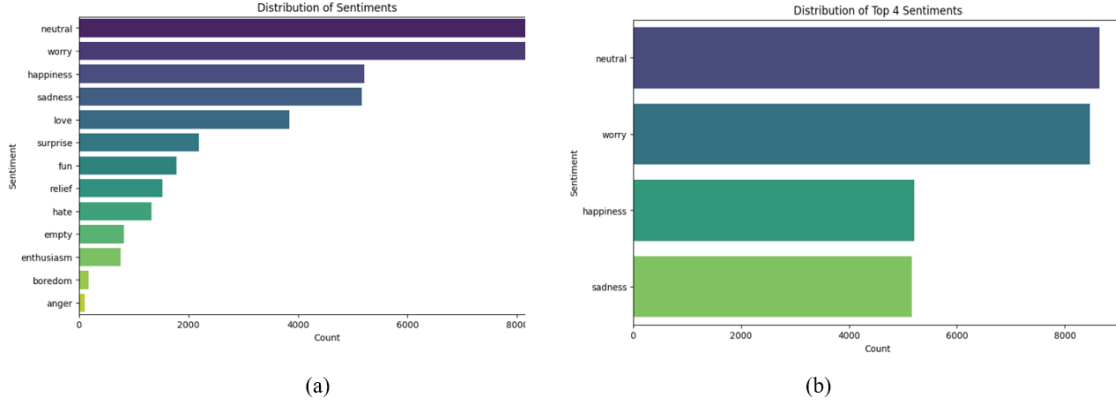


Fig. 2: Distribution of Tweet emotion dataset: (a) all labeled emotions and (b) mapped basic emotions.

Table 2: Sample of Tweet Emotion Dataset

Tweet ID	Content / Tweet	Sentiment
1956967666	Layin n bed with a headache ugh-hhh...waitin on your call...	Sadness
1956967789	Wants to hang out with friends SOON!	Happiness
1956968416	@dannycastillo We want to trade with someone who has Houston tickets, but no one will.	Neutral
1753919001	Happy Mother's Day All my love	Happiness

To test the proposed machine learning system, a book review dataset consisting of approximately 300,000 reviews was used [23]. These reviews were collected from users across various books and also include user-provided ratings. Using Google's Book API, the system predicts the sentiment embedded in these reviews to build an intelligent recommendation engine based on emotional alignment. A sample of this dataset is shown in Table 3.

Table 3: Sample data from the Book Review Dataset

Book ID	User ID	Book Title	Review Description	Rating
1847022251	AU6DIIDZK2OQM	Jane Eyre (LargePrint)	Jane was orphaned as a baby and thrust upon her... love almost lose	4.0
0435126024	AU6DIIDZK2OQM	Jane Eyre (New Windmill)	Jane was orphaned as a baby and thrust upon her... love almost lose	4.0
B000ILIJEO	AU6DIIDZK2OQM	The Hobbit There and Back Again	Bilbo Baggins is a Hobbit, a creature unlike a... mind numb	1.0
0192824325	AU6DIIDZK2OQM	Emma (The World's Classics)	Emma is twenty years old, living in the country... shallow live	2.0

3.2 Data Preprocessing

Data preprocessing is an essential step in natural language processing (NLP), aimed at transforming raw text data into a clean and structured format suitable for analysis and modeling [24]. Effective preprocessing enhances model performance by mitigating issues such as noise, irrelevant information, and inconsistencies, which are common in real-world datasets. Given that the tweet dataset was collected from crowdsourced sources, it contains considerable noise, necessitating comprehensive preprocessing. To show the working process let us take a sample sentence from the Tweet Emotion Dataset: *@tiffanylue i know i was listenin to bad habit earlier and i started freakin at his part* =['

A. Tokenization: Tokenization splits raw text into individual units called tokens, typically words or punctuation marks, which serve as the foundation for further analysis [25]. This study utilizes the Natural Language Toolkit (NLTK) library for tokenization. Tokenizing the example sentence yields: ['@tiffanylue', 'i', 'know', 'i', 'was', 'listenin', 'to', 'bad', 'habit', 'earlier', 'and', 'i', 'started', 'freakin', 'at', 'his', 'part', '=['

B. Strip HTML Parser: A parser, in the context of web scraping, web harvesting, or web data extraction is data scraping used for extracting data from websites [26]. The purpose of a parser is to analyze the document's markup language (such as HTML or XML) and convert it into a more structured and usable format. Parsers are commonly used to navigate through the hierarchy of tags and elements in a document, allowing developers to extract specific data or perform various manipulations. An HTML parser is a program or library that reads HTML code and translates it into a more structured format, often represented as a tree-like structure called the parse tree or document object model (DOM). In this case, BeautifulSoup is a popular Python library that provides tools for pulling data out of HTML and XML files. It creates a parse tree from the page's source code, which can then be navigated and searched to extract the required information. Example: `<title>Example Page</title>` is a HTML snippet. Using BeautifulSoup, a parse tree is created and elements can be accessed by `soup = BeautifulSoup(html, 'html.parser')` and then `title = soup.title`.

In the sample sentence, there are no HTML tags. So, the tokenized parts would remain unchanged: ['@tiffany- lue', 'i', 'know', 'i', 'was', 'listenin', 'to', 'bad', 'habit', 'earlier', 'and', 'i', 'started', 'freakin', 'at', 'his', 'part', '=['

C. Remove Non-ASCII Characters and Punctuation: As the name suggests, the python libraries are used to remove non-ascii characters and punctuations from the tokenized words [27]. The sample sentence will be preprocessed here and the output will be: ['tiffanylue', 'i', 'know', 'i', 'was', 'listenin', 'to', 'bad', 'habit', 'earlier', 'and', 'i', 'started', 'freakin', 'at', 'his', 'part'].

D. Replace Contractions: Then used another python library named contractions to remove contractions and use the extended words [28]. For example: if there are any didn't or don't in the sentence they are expanded to show as didnot or donot. The sample sentences contractions will be removed and will show like this: ['tiffanylue', 'i', 'know', 'i', 'was', 'listenin', 'to', 'bad', 'habit', 'earlier', 'and', 'i', 'started', 'freakin', 'at', 'his', 'part'].

E. Replace Numbers: Consequently, python library inflect is used to replace all integer occurrences in the list of tokenized words with textual representation. For example, in this tokenized sentence ['There', 'are', '5', 'reasons', 'for', 'this.'], there is numeric text. In text analysis converting the number 5 to a text would be more human-friendly and linguistically natural [29]. So, after preprocessing this tokenized part turned into ['there', 'are', 'five', 'reason', 'for', 'this']. Since there are no numbers in the sample sentence the tokenized words will remain the same. The unchanged output will be: ['tiffanylue', 'i', 'know', 'i', 'was', 'listenin', 'to', 'bad', 'habit', 'earlier', 'and', 'i', 'started', 'freakin', 'at', 'his', 'part'].

F. Lemmatization: Lemmatization is a text preprocessing technique in natural language processing (NLP) that involves reducing words to their base or canonical form, known as the lemma. The primary goal of lemmatization is to standardize words, ensuring that different inflected forms of a word are represented uniformly [30]. This process aids in improving the efficiency and accuracy of various NLP tasks by reducing words to their most basic and meaningful forms. Lemmatization combines of two forms these are given below:

1. Retaining meaningful base forms: Lemmatization aims to reduce words to their meaningful base forms or lemmas. This helps in preserving the semantic meaning of words. Such as, ‘running’ → ‘run’, ‘happily’ → ‘happy’.

2. Handling irregular forms: Lemmatization is capable of handling irregular words that don’t follow regular inflectional patterns. It considers the word’s part of speech and provides a more accurate base form. Instance of, ‘children’ → ‘child’, ‘went’ → ‘go’.

After Lemmatization, the sample word will look like this: [‘tiffanylue’, ‘i’, ‘know’, ‘i’, ‘wa’, ‘listenin’, ‘to’, ‘bad’, ‘habit’, ‘earlier’, ‘and’, ‘i’, ‘start’, ‘freakin’, ‘at’, ‘his’, ‘part’]

G. Remove Stopwords: It refers to the process of eliminating common words, known as stopwords, from a piece of text. Stopwords are words that are frequently used in a language but typically don’t carry significant meaning on their own [31]. Examples of stopwords in English include words like ‘the’, ‘and’, ‘is’, ‘in’, and so on. After this process, the sample sentence will be looking like this: [‘tiffany- lue’, ‘know’, ‘listenin’, ‘bad’, ‘habit’, ‘earlier’, ‘start’, ‘freakin’, ‘part’].

H. Normalization: Normalization is a preprocessing step that scales and transforms numerical features to a standard range. It ensures that different features contribute equally to the analysis and prevents certain features from dominating due to their larger magnitudes. Normalization is particularly important in machine learning algorithms that are sensitive to the scale of input features, such as gradient-based optimization algorithms [32]. Lowercasing, removing HTML parser, non-ASCII characters, punctuations, lemmatizations are some examples of normalization. So, the final output will look like this: [‘tiffanylue’, ‘know’, ‘listenin’, ‘bad’, ‘habit’, ‘earlier’, ‘start’, ‘freakin’, ‘part’].

To evaluate the data preprocessing steps whether both the datasets are preprocessed correctly, extracted some contents from the datasets and output the result after preprocessing. From the Table 4 shows the before and after preprocessing condition of the sampled texts from Tweet emotion dataset.

Table 4: Before and after process of tweet emotion data

Before Text Preprocessing	After Text Preprocessing
@tiffanylue i know i was listenin to bad habit ear- lier and i started freakin at his part =[	tiffanylue know listenin bad habit earlier start freakin part
Layin n bed with a headache ughhhh...waitinon your call...	layin n bed headache ughhhh waitin call
wants to hang out with friends SOON!	want hang friends soon

3.3 Feature Extraction

TF-IDF [33] is a numerical representation technique commonly used in natural language processing and feature extraction. It evaluates the importance of a word in a document relative to a collection of documents, considering both the term frequency (TF) and the inverse document frequency (IDF). The TF-IDF follows three steps to extract the feature and these are:

- 1. Term Frequency (TF):** Measures how frequently a word appears in a document and the mathematical equation for TF in “(1)”

$$tf(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d} \quad (1)$$

- 2. Inverse Document Frequency (IDF):** Highlights the rarity of a term across the entire document collection and as calculate by “(2)”:

$$idf(t, D) = \log \frac{\text{Total num of documents in the collection}}{\text{Num of docs containing term } t + 1} \quad (2)$$

- 3. TF-IDF Score:** TF-IDF score is calculated using the formula “(3)”:

$$tf_idf(t, d, D) = tf(t, d) \times idf(t, D) \quad (3)$$

Where, t is the term or word, d is the single document, D is the document corpus or collection. Now, given a calculation for TF-IDF on sample sentence of dataset.

- **Document-1:** “This is such a great book”
- **Document-2:** “I love this book”
- **Term Frequency (TF):** *Count of “great” in Document-1 = 1, Total number of words in Document-1 = 6*
- **TF (“great”, Document 1) = 1/6**
- **Inverse Document Frequency (IDF):** *Total number of documents = 2 Number of documents containing “great” = 1* $IDF(\text{“great”}) = \log(2) = 0.301$
- **TF-IDF Calculation:** $TF-IDF(\text{“great”, Document-1}) = TF(\text{“great”, Document-1}) \times IDF(\text{“great”})$
 $TF-IDF(\text{“great”, Document-1}) = \frac{1}{6} \times 0.301 \approx 0.0502$

Therefore, the TF-IDF score for the term “great” in Document-1 is approximately 0.0502. This score indicates the relative importance of the term “great” within Document-1 compared to the entire document collection.

3.4 Data Balancing

By adding significantly modified copies of either existing data or newly created synthetic data from existing data or removing noisy redundant data from the sample dataset, techniques are employed to expand the amount of data. These techniques are called Data Balancing. To resolve the class imbalance in the data, this study used two data balancing methods and these are ADASYN (Adaptive Synthetic) and ENN (Edited Nearest Neighbor).

1) ADASYN: ADASYN (Adaptive Synthetic) [34] is an oversampling method used in the field of machine learning. Imbalanced datasets, where one class is underrepresented, can lead to biased models favoring the majority class. ADASYN addresses this challenge by dynamically generating synthetic samples for the minority class. ADASYN follows three steps to sampling data, which given below:

Step-1: Calculate the imbalance ratio:

$$IR(c) = \frac{N_{Majority}}{N_{Minority}} \quad (1)$$

Step-2: Calculate the desired number of synthetic Instance:

$$G(c) = ID(c) \times (\text{Number of majority instances}) - 1 \quad (2)$$

Step-3: Generate Synthetic Instances: For each minority class instance:

- Calculate the Euclidean distance to its k nearest neighbors in the feature space
- Normalize the distances to obtain a density distribution.
- Calculate the density ratio for each minority instance:

$$D(c) = \frac{\text{density of minority instance}}{\sum_{i=1}^k \text{density of neighbor}_i} \quad (3)$$

- Generate $G(c)$ synthetic instances around the minority instance based on the normalized density ratios.

2) ENN: ENN (Edited Nearest Neighbors) [35] is an under-sampling method used in the context of addressing imbalanced datasets in machine learning. Under sampling technique aim to balance class distribution by reducing the number of instances in the over-represented class, typically the majority class, so that it is more proportionate to the minority class. In the ENN algorithm, each instance in the dataset is examined, and its class label is verified by considering its k -nearest neighbors. If an instance is found to be misclassified based on a majority vote from its neighbors, it is considered noisy and is subsequently removed from the dataset. ENN operates as a data cleaning step, intending to improve the robustness and generalization of a model. By eliminating instances that are likely to be mislabeled or outliers, ENN

contributes to creating a more balanced and reliable training set. ENN can be described as below:

Step-1: Identify Misclassified Instances: For each majority class instance x_i with label $y_i = 0$ (assuming binary classification), find its k nearest neighbors N_i in the majority class:

$$N_i = \underset{j \text{ s.t. } y_j=0}{\operatorname{argmin}} \|x_i - x_j\| \quad (4)$$

Identify instances x_i where the majority class label of N_i is different from y_i , and mark them for removal.

Step-2: Remove Marked Instances: Remove the instances marked for removal from the majority class.

For the split of 90% for training and 10% testing, data preprocessing was performed. For the training and evaluation of the transformer models, the dataset was partitioned into three distinct subsets: 80% was allocated for the training set, 10% for the validation set, and the remaining 10% for the test set. This included the reduction of noise and elimination of mislabeled and redundant samples completed by using Edited Nearest Neighbor (ENN) from the original dataset as part of the preprocessing. This data reduction has an influence on the model performance as can be seen in the confusion matrix in Figure 6. The preprocessing will make sure that the model is trained on better data that is representative of the selected question. This potentially leads to an improved classification accuracy and less errors made by the model.

3.5 Model Implementation

To develop the book recommend system, three machine learning models are employed with stratified K-fold approach to train the dataset. Data normalization is used in machine learning and statistics to scale and transform the features of a dataset. The goal is to bring the features to a similar scale or range. The description of model is given below:

A. Random Forest (RF)

Random Forest [36] is a powerful and versatile ensemble learning method widely used in machine learning for both classification and regression tasks. It belongs to the family of ensemble methods, which combine the predictions of multiple individual models to achieve better overall performance. The equation for random forest is defined in Equation 5 as:

$$\hat{y} = \operatorname{model}(f_1(x), f_2(x), \dots, f_n(x)) \quad (5)$$

Where, $\hat{y}$ is the predicted output, $f_1(x), f_2(x), \dots, f_n(x)$ are the predictions of individual decision trees for the input feature vector x . The function *model* represents the most frequently occurring prediction (mode).

In random forest, `balanced_subsample` is used as class weight because it is used to automatically adjust the weights of classes during the training phase. The criterion is set to entropy (information gain). It measures the information contained in each node. And `verbose` was set to 2 because it outputs information about each tree, including the progress bar.

B. Support Vector Machine (SVM)

Support Vector Machine (SVM) [37] is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the best possible line or hyperplane that separates different classes in the data with the largest possible margin. SVM aims to maximize the margin between classes while minimizing classification errors. It can handle both linear and non-linear data by using a technique called the kernel trick to map data into higher-dimensional spaces. The SVM decision function equation in Equation 6 is defined as:

$$f(x) = w \cdot x + b \quad (6)$$

Where, w is the weight vector, x is the input vector and b is the bias term.

C. Logistic Regression (LR)

Logistic Regression [38] is a statistical method used for binary classification problems, where the outcome variable is categorical and has only two possible values, typically denoted as 0 and 1. It's a part of the broader family of regression analysis, but it is more suited for classification tasks. The logistic regression model predicts the probability that a given instance belongs to a particular category. In logistic regression, the logistic function (also called the sigmoid function) is used to model the probability of a binary outcome. The logistic function is defined in Equation 7 as:

$$g(z) = \frac{1}{1 + e^{-z}} \quad (7)$$

Where, $g(z)$ represents the output of the logistic function, e is the base of the natural logarithm, and z is the input to the function, typically a linear combination of features in logistic regression.

D. Stratified K-Fold

Stratified K-Fold Cross-Validation [39] is a variation of K-Fold Cross-Validation, a common technique used in machine learning to assess the performance of a model. The stratified version ensures that each fold preserves the percentage of samples for each class, which is particularly useful when dealing with imbalanced datasets. The equation for Stratified K-Fold is defined as:

$$D_k = D_{k=1}^K D_E \quad (8)$$

$$D_k = \{(X_{k1}, y_{k1}), (X_{k2}, y_{k2}), \dots, (X_{kn}, y_{kn})\} \quad (9)$$

Where, D is the dataset, k is the number of folds or partitions in the cross-validation process, and (X_{k1}, y_{k1}) is each element in the dataset with a pair of features and labels.

For stratified K-fold, 10-fold is used and dataset is divided into 10 folds. The models will be trained and evaluated 10 times, each time using a different fold as a validation set and the remaining others as training sets. It helps the models not to be dependent on a specific set of data.

E. DistilBERT

DistilBERT is a smaller, faster, and lighter version of BERT created using knowledge distillation. Instead of training from scratch, a student model (DistilBERT) learns to mimic a larger teacher model (BERT). The training objective combines three losses:

$$L = \alpha \cdot L_{\text{MLM}} + \beta \cdot L_{\text{distill}} + \gamma \cdot L_{\text{cosine}} \quad (10)$$

Where L_{MLM} is the masked language modeling loss, L_{distill} minimizes the Kullback–Leibler (KL) divergence between the teacher and student logits, and L_{cosine} aligns the hidden states. DistilBERT reduces parameters by approximately 40%, runs about 60% faster, and retains nearly 97% of BERT's performance.

F. BERTweet

BERTweet, in contrast, is a transformer-based language model pretrained on 850M English tweets. Because Twitter data includes slang, hashtags, and informal grammatical structures, BERTweet is designed to capture these language constructs better than general-purpose models like BERT. Like RoBERTa, BERTweet is pretrained with the same objectives: Masked Language Modeling (MLM). BERTweet represents significant improvements specifically in social media-based sentiment analysis, offensive language detection, and misinformation classification.

G. XLM-RoBERTa

XLM-RoBERTa is an extension of RoBERTa trained on 2.5TB of filtered CommonCrawl data in 100 languages using the Masked Language Modeling (MLM) objective. It does not rely on language-specific segmentation like mBERT but uses SentencePiece tokenization, enabling robust cross-lingual generalization. XLM-RoBERTa achieves strong results on cross-lingual benchmarks (e.g., XNLI, MLQA) and is widely used for multilingual NER, MT, and sentence retrieval.

H. Model Fine-Tuning

Fine-tuning a model is a transfer learning process where the goal is to train the pre-trained model on a target dataset, in order to customize the model’s weights for a specific task. Instead of training a model from zero, fine-tuning utilizes the knowledge already gained from training on large datasets that uses less time when training the model while improving performance given that they often have limited data. As an example, in NLP, there are models such as BERT or XLM-Roberta which are pre-trained on larger quantities of text data, which can then be fine-tuned etc. on a labeled dataset, to detect sentiment or emotion classification.

The use of classical machine learning algorithms like Support Vector Machine, Random Forest, and Logistic Regression is driven by practical as well as theoretical reasons. These algorithms are more interpretable than complicated deep learning models, enabling a better understanding and explanation of decision-making processes—a consideration that is crucial for establishing trust and transparency in recommender systems. Furthermore, classical algorithms require much fewer computational resources, making them easier to deploy in hardware-constrained environments or where real-time processing is necessary. Coupled with the right strategies like data balancing using ADASYN and robust assessment via stratified K-fold cross-validation, these algorithms showcase strong and stable performance on imbalanced data, as demonstrated by the high accuracy obtained in emotion prediction. This trade-off of efficiency, interpretability, and predictive capability warrants the methodological choice of classical machine learning algorithms in this research.

4 Results and Discussions

For training the Tweet emotion dataset, the 80:20 ratio approach is used that is meaning 80% data of the dataset is used for training and the remaining 20% used for validation purposes. Also, the mapped 5 emotions are converted to numeric labels which are used as Y datapoints and the text contents are normalized which are used as X datapoints.

To evaluate our sentiment prediction model, though accuracy is the most commonly used for performance matrices, to get a better understanding several other performance metrics is computed, such precision, and F1-score for the evaluation of the proposed system and their equations is presents in “(11)” to “(15)”.

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{All the sample of Sentiment}} \quad (11)$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (12)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (13)$$

$$F1 \text{ Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

$$\text{Specificity} = \frac{\text{True Negative}}{\text{False Positive} + \text{True Negative}} \quad (15)$$

The evaluation of the proposed system’s performance encompasses various metrics to enhance the effectiveness and efficiency of the book recommendation system. These metrics include assessing the model’s performance without implementing stratified k-fold validation, and stratified k-fold validation with varying values of k (k= 1 to 10). The outcomes of these evaluations are summarized below:

A. Model performance without fine-tuning:

Table 5 shows the relative performance of traditional machine learning models in comparison to transformer-based models without fine-tuning. Random Forest performed the best among the machine learning models, with an accuracy of 84.83% and an F1 score of 84.00%, followed by Support Vector Machines (SVM) and Logistic Regression, which reported lower but still reasonable performance. By contrast, the transformer models (BERTweet, XLM-RoBERTa, and DistilBERT) performed significantly worse, with

accuracies between 52% and 54% and F1 scores around 53%. This is indicative of the fact that, while transformer models are powerful for general natural language processing tasks, they perform poorly when applied directly without fine-tuning since they do not adequately adapt to the unique properties of the dataset. In contrast, traditional machine learning models had better performance in this specific task due to their ability to learn optimally from the dataset without requiring large-scale pretraining or domain adaptation.

Table 5: Performance Evaluation of ML & NLP models without fine-tuning and transformer models

Model	Accuracy	Precision	Recall	F1 Score
RF	84.83%	90.00%	81.00%	84.00%
SVM	76.60%	84.00%	68.00%	74.00%
LR	74.94%	84.00%	54.00%	61.00%
BERTweet	53.71%	53.72%	55.77%	53.96%
XLM-RoBERTa base	53.31%	53.16%	53.77%	53.28%
DistilBERT	52.69%	52.23%	52.78%	52.47%

B. Model performance with stratified k-fold/fine-tuning:

Table 6 compares the performance of machine learning models with fine-tuning and transformer models. Following fine-tuning, the conventional ML models showed additional improvement. SVM performed the best with 87.47% accuracy and 88.00% F1 score, closely followed by Random Forest with 85.43% accuracy and 86.00% F1 score. Logistic Regression also performed better, achieving 79.44% accuracy and 81.00% F1 score. These findings accentuate that fine-tuning maximized the learning potential of ML models, and they became more effective in processing the classification task. Conversely, the transformer models (BERTweet, XLM-RoBERTa, and DistilBERT) once again exhibited very low performance, with around 52–54% accuracy and 53% F1 scores, similar to the findings in Table 5. This further confirms that transformer models need task-specific fine-tuning on the dataset to become competitive and, when left unfine-tuned, cannot compete with the effectiveness of conventional ML models in this research.

Table 6: Performance Evaluation of ML & NLP models with fine-tuning and transformer models.

Model	Accuracy	Precision	Recall	F1 Score	Specificity
SVM	87.47%	94.00%	84.00%	88.00%	
RF	85.43%	90.00%	83.00%	86.00%	
LR	79.44%	87.00%	77.00%	81.00%	
BERTweet	53.71%	53.72%	55.77%	53.96%	
XLM-RoBERTa base	53.31%	53.16%	53.77%	53.28%	
DistilBERT	52.69%	52.23%	52.78%	52.47%	

A comparison was conducted between a standard Logistic Regression model and a fine-tuned version, on a dataset of 3,137 samples. The standard model correctly classified 2,351 samples; however, its performance was greatly hindered by confusion between the neutral and worry classes, with the greatest error being the misclassification of 418 true worry instances as neutral. This model struggled with the happiness class, with correct identifications for only 92 of 227 samples, and underperformed on the sadness class, with only 4 correct predictions out of 16. By contrast, the fine-tuned Logistic Regression model showed a dramatic improvement, with correct classification of a total of 2,492 samples. In this model, the main source of error shifted to the confusion between the happiness and neutral classes, with 409 true neutral instances incorrectly predicted as happiness. Despite this change, the fine-tuned model showed substantially better performance on the sadness class, with 191 correct classifications.

In fig. 4, a comparison was made between a baseline SVM and a fine-tuned SVM on a data set of 3,137 samples. The baseline SVM classified 2,403 samples correctly. Its greatest weakness was the confusion between the neutral and worry classes, exemplified by its largest single mistake: 401 true worry instances predicted as neutral. This model identified 1,497 neutral instances and 755 worry instances correctly, and for the happiness class, it classified 142 out of 227 samples correctly. By comparison, the fine-tuned SVM showed a dramatic improvement, classifying a total of 2,744 samples correctly. This model

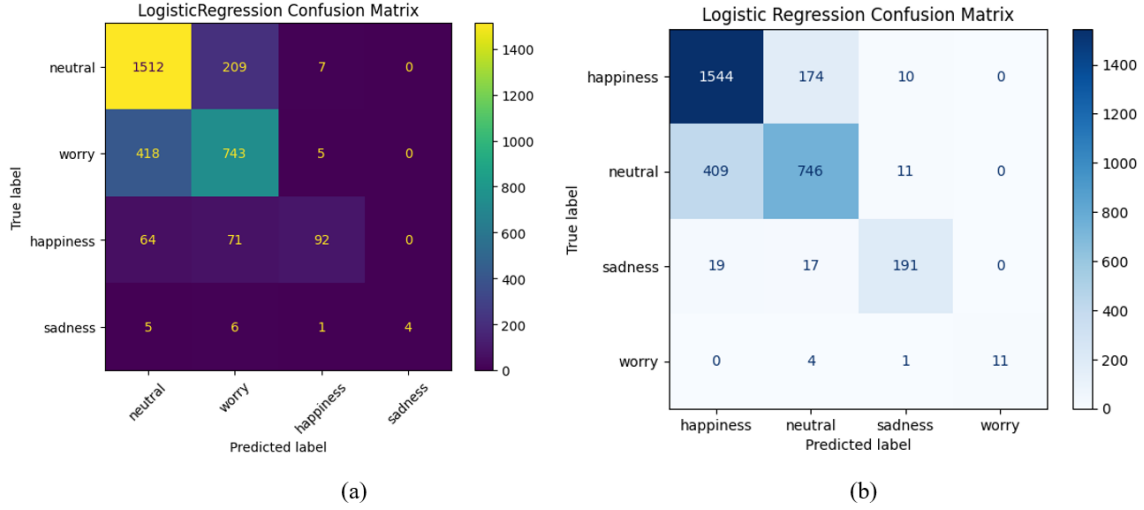


Fig. 3: Confusion matrix of Logistic Regression (a) without fine-tune and (b) with fine-tune validation

performed almost flawlessly for the happiness and sadness classes, identifying 1,655 and 222 instances correctly, respectively. However, its main source of error changed; the largest misclassification was 307 true neutral instances predicted as happiness, which kept the number of correct neutral predictions down to 856. Although the fine-tuning procedure improved the number of correct classifications overall, it also changed the error profile, eliminating some but not all of the worry-neutral confusion but introducing a novel systematic error between the neutral and happiness classes.

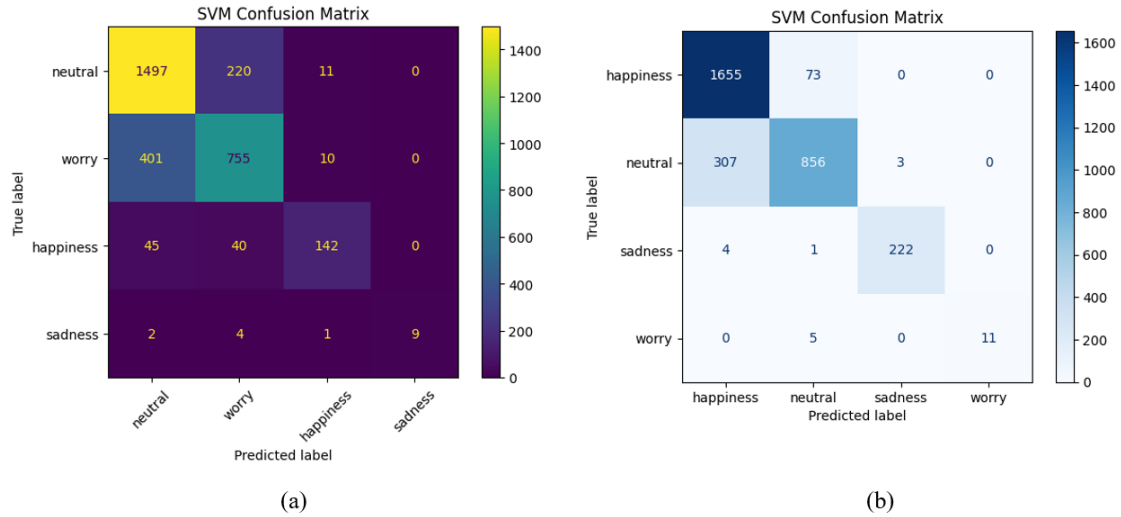


Fig. 4: Confusion matrix of Support Vector Machine (a) without fine-tune and (b) with fine-tune validation.

In fig. 5, a comparison of a baseline Random Forest model and a fine-tuned alternative was performed on a data set of 3,137 samples. The baseline model correctly predicted 2,661 samples, with particular success at predicting the happiness class with 221 correct predictions out of 227. Its most noticeable mistake, however, was the misclassification of 382 instances of worry as neutral, restricting the correct predictions for worry to 779 out of 1,166. The fine-tuned model, on the other hand, performed a slightly greater number of correct classifications at 2,680. This model performed particularly well with the happiness and sadness classes, predicting 1,649 and 221 instances correctly, respectively. Its main weakness was reversed for the misclassification between neutral and happiness, in which 360 actual neutral instances

were mis predicted as happiness, leaving only 799 correct predictions for the neutral class. Though the fine-tuned model had a slightly greater number of overall correct predictions, both models exhibited unique error patterns, with the baseline model confusing worry with neutral and the fine-tuned model confusing neutral with happiness.

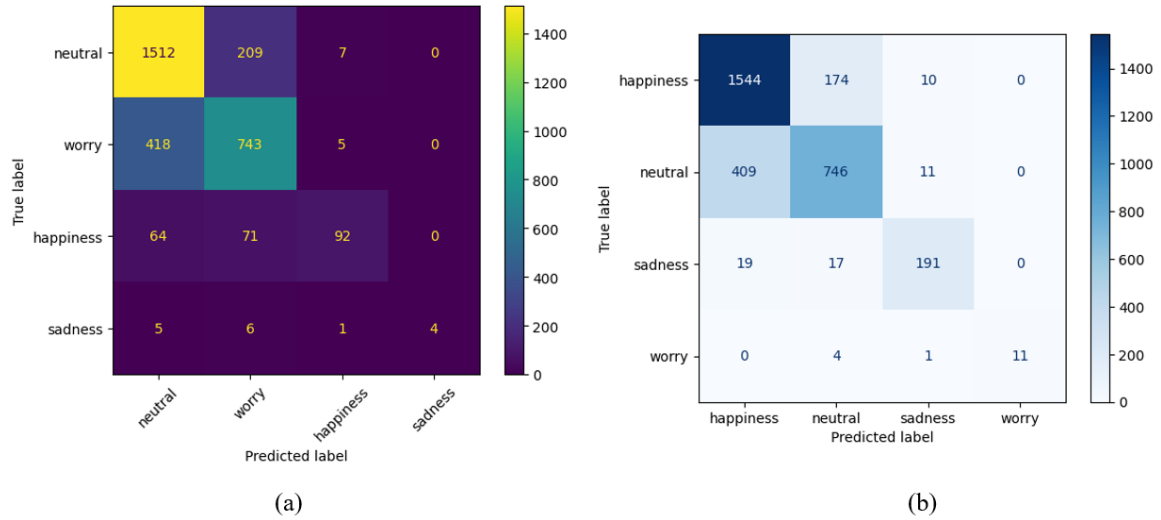


Fig. 5: Confusion matrix of Random Forest: (a) without fine-tuning and (b) with fine-tuning validation.

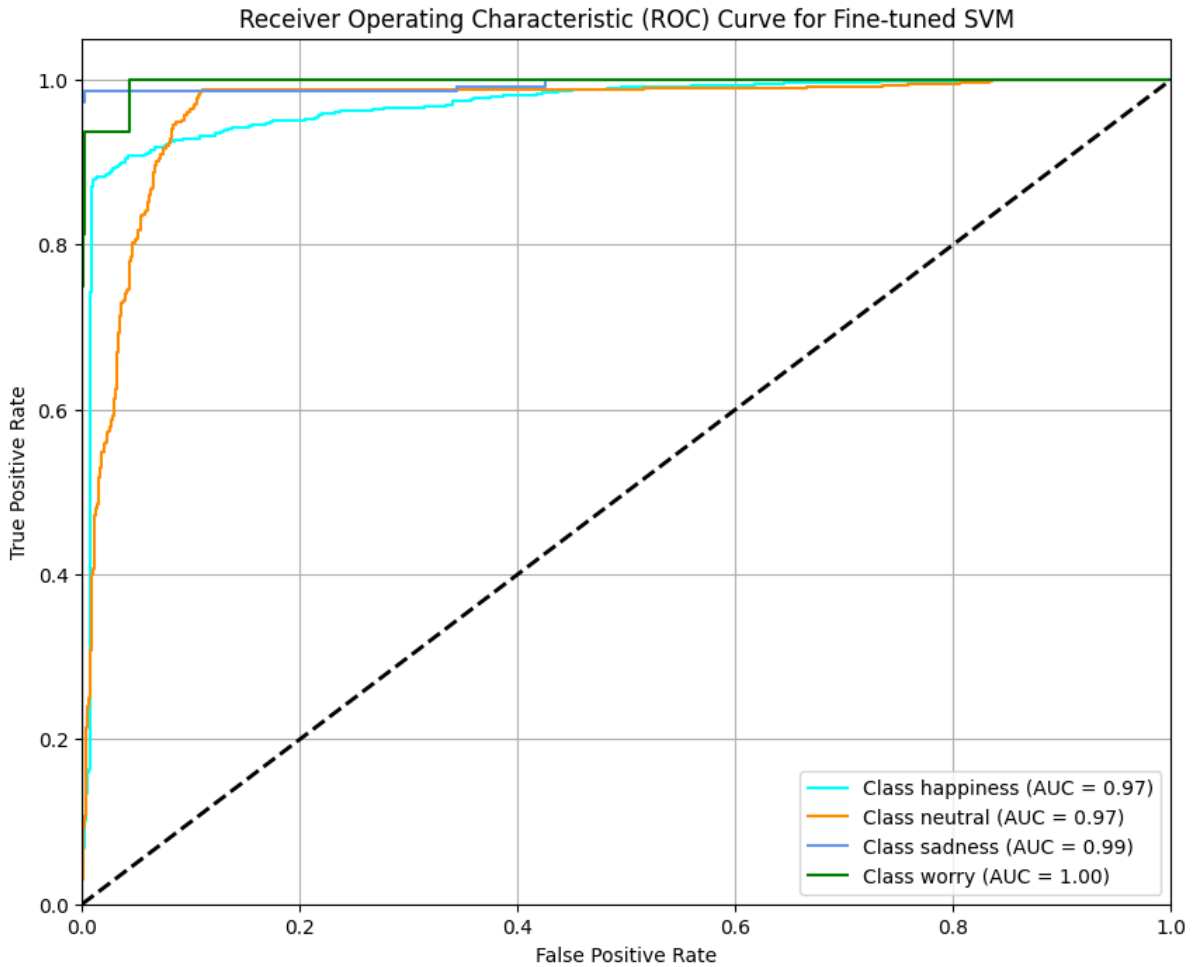


Fig. 6: ROC Curve from SVM Fine-Tuning.

The ROC plot displays the performance of the fine-tuned SVM model over four classes of emotions: happiness, neutral, sadness, and worry. Each curve is the trade-off between true positive rate and false positive rate for a class, and the respective Area Under the Curve (AUC) values measure the discriminative capacity of the model. The performance is seen to be very strong classification, with happiness and neutral scoring AUC values of 0.97, sadness scoring an AUC of 0.99, and worry obtaining a perfect score of 1.00. These high AUC values express the strength of the fine-tuned SVM in separating the emotional classes with very little misclassification. To check if the sentiment analysis model is overfit or not, then introduce a learning curve for training and validation data. From Figure 8, the training curve is almost 100 from the start. In this case, it is clam that the model is learning very well. Even though it can be said as overfitted if we look at the validation learning curve, it is indicating a steady climb from start to end, which makes the impression that the model is a good fit. If the model is overfitted for the unseen data in the validation part it would rise sharply or drop significantly and if the model is underfitted, it would maintain a low curve but here showing a steady rise till the end.

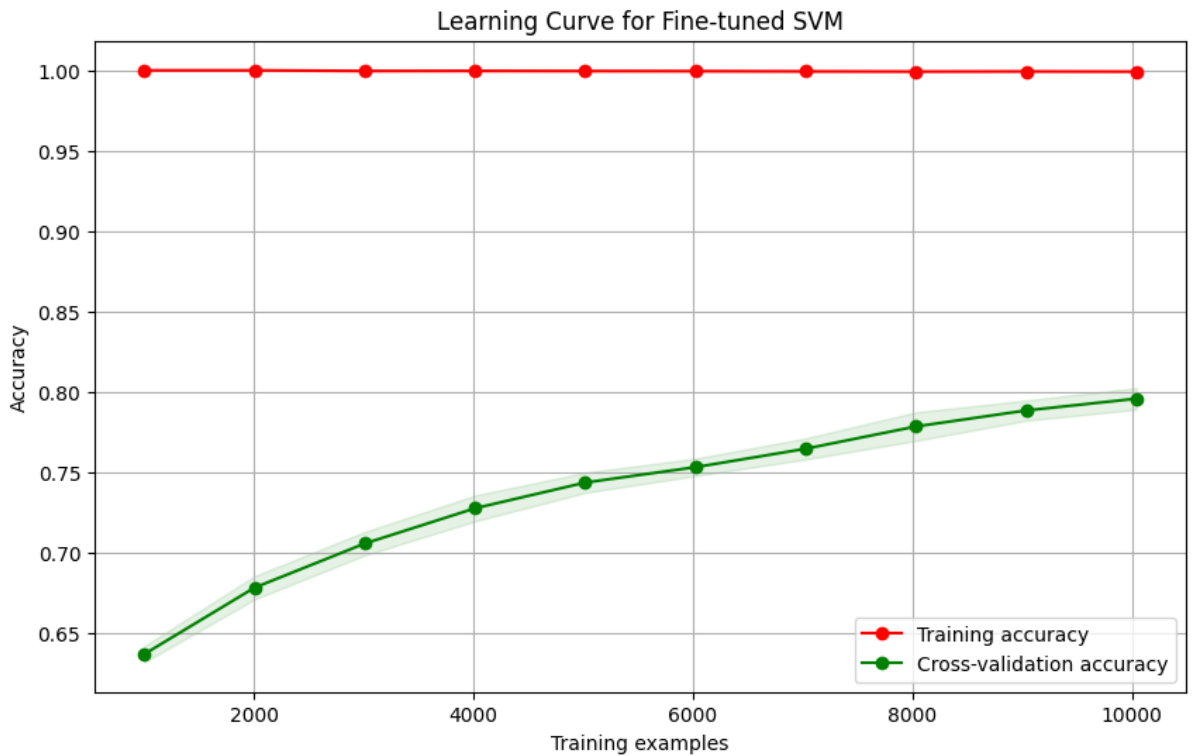


Fig. 7: Learning curve of SVM with stratified K-fold.

A comparison study of three transformer-based models—XLM-RoBERTa, DistilBERT, and vinai/bertweet-base—on a test dataset of 2,748 samples was performed to compare their performance. Overall accuracy was similar for all three models but with differing error patterns. As outlined in the confusion matrices in Figure 7, the XLM-RoBERTa model accurately classified 1,465 samples, with its best performance on the worry (457) and neutral (430) classes. Its biggest weakness was confusing other emotions for worry, with its highest single mistake being 207 instances of true sadness predicted incorrectly as worry. The DistilBERT model, also in Figure 7, accurately identified 1,448 samples and likewise had confusion between sadness and worry. Its most common mistake was incorrectly labeling 195 true instances of worry as sadness, reflecting a notable overlap in how both models perceive these negative affective states. The vinai/bertweet-base model, the performance of which is shown in the confusion matrix in Figure 8, had the best accuracy among the three by accurately predicting 1,476 samples. Its best-performing class was neutral (446 correct classifications), but similar to the other two models, was brought down by considerable confusion between semantically close emotions. Its largest mistake was wrongly classifying 258 actual instances of worry as sadness.

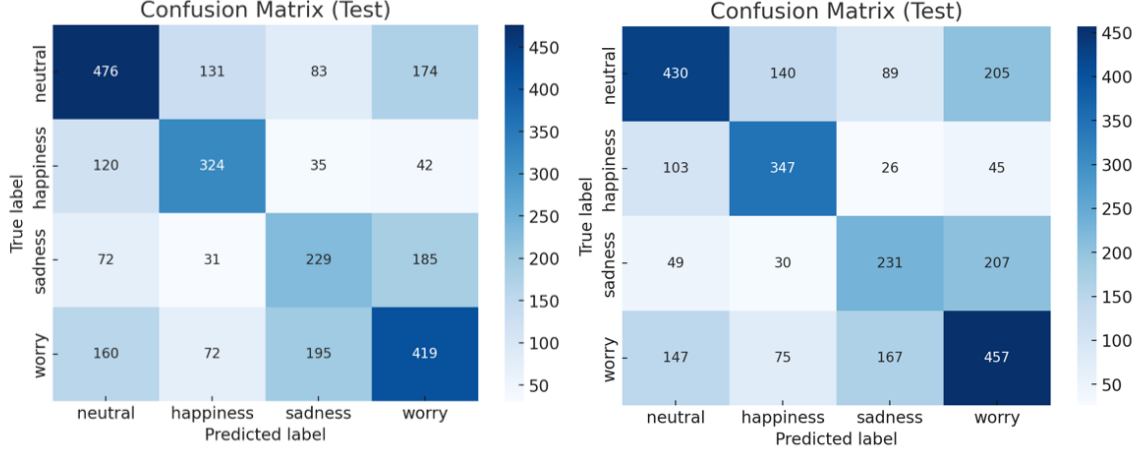


Fig. 8: Confusion matrix of DistilBERT, and XLM-RoBERTa base.

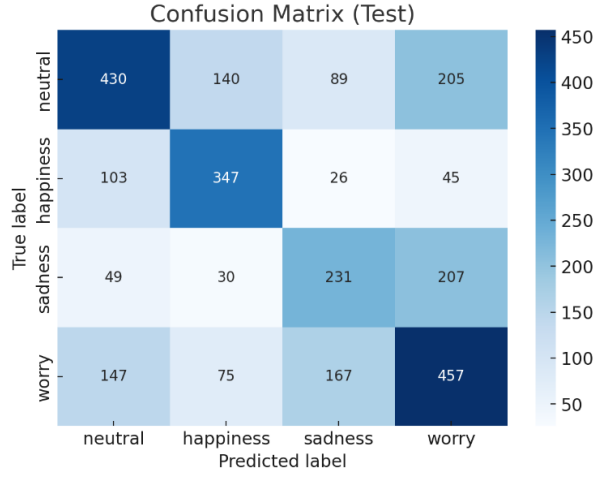


Fig. 9: Confusion matrix of BERTweet.

4.1 Predicting Emotions and Genre

By using the SVM with stratified K-fold model, we predict the emotions from the Tweet Emotion Dataset and also the Book Review Dataset that are used.

The proposed methodology for genre prediction involved leveraging Google’s Book API, which boasts a comprehensive database comprising over 40 million books across more than 500 languages. Utilizing this API, the genre predictions are conducted by inputting specific book titles from the dataset [42]. For instance, when querying a renowned title such as ‘Romeo and Juliet’ the API provided a genre classification that encompassed categories such as [‘Children’s plays’, ‘Drama’, ‘Vendetta’, ‘One-act plays, American’, ‘Fiction’]. This process allowed us to assign genres to the books within the dataset based on the information retrieved from the Google Books API.

4.2 Book Recommendation System

To improve the sentiment-based book recommendation system, refined the raw review data by including only users with over 200 reviews and books with at least 50 reviews. We filtered out reviews with minimal text and simplified the emotion categories into Positive, Negative, and Neutral. Using a trained SVM model, we analyzed a user’s past reviews for different book genres to recommend or avoid genres based on their overall sentiment. If the reviews were mostly positive, we recommended books from that genre, and if they were negative, we avoided suggesting that genre [43,44]. When positive and negative reviews were equal, the system resulted in a neutral outcome, indicating mixed feelings toward that genre.

And also added on how the recommendation model is recommending book to a user. From Figure ?? (a), that positive review on a genre is much higher than negative and neutral. That's why the recommendation system will recommend that genre's book to a user. And Figure ?? (b) shows the opposite when a genre's book will not be recommended.

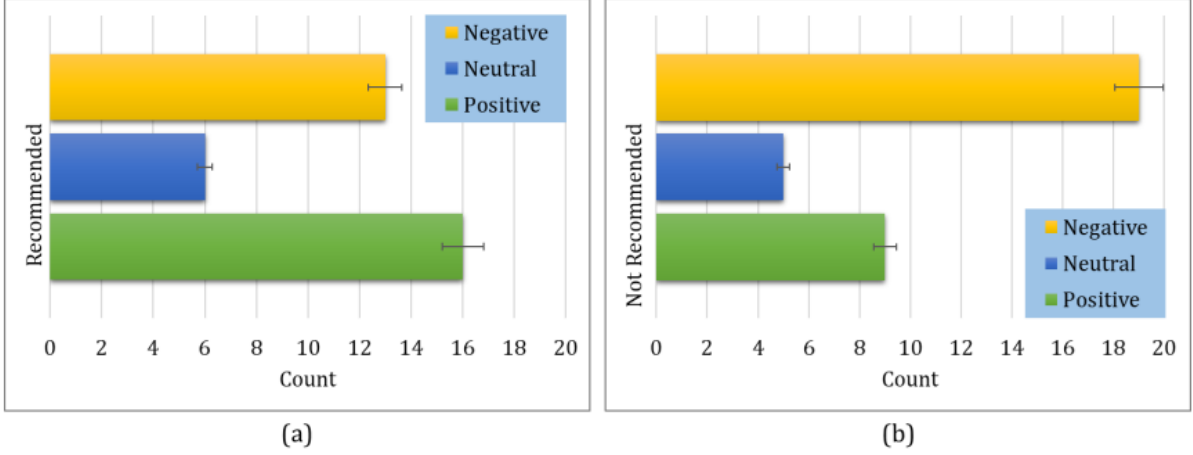


Fig. 10: The diagram for book (a) recommended to user and (b) not recommended to user.

5 Discussions

The implementation of data balancing methods, such as ADASYN and ENN, has significantly contributed to enhancing the performance of the proposed system. As previously highlighted, the Tweet emotion dataset exhibits class imbalance, leading to suboptimal accuracy and biased results when employing machine learning models. Table 7 illustrates that, prior to the application of data balancing methods, the accuracy of all machine learning models was below 50%, underscoring the detrimental impact of the dataset's high skewness. In the process of balancing the data, ADASYN effectively addresses class imbalance by resampling all classes, introducing additional data points with slight variations to the minority classes. On the other hand, ENN focuses on the majority classes, eliminating data points by considering the consensus of nearest neighbors. If there is unanimous agreement among neighbors on the sample's class label, the sample is retained, but if disagreement arises, indicating a potentially ambiguous region, the sample is excluded.

However, it is noted that the performance of ENN was deemed suboptimal, leading to its exclusion from further consideration. Subsequently, Table 7 showcases a marked improvement in the performance of machine learning models post the application of data balancing techniques. These methods have proven instrumental in mitigating the challenges posed by class imbalance, resulting in a significant boost in the overall performance of the considered machine learning models.

Table 7: Model accuracy before and after fine-tuning.

Model Name	Accuracy Before	Accuracy After
Random Forest	84.83%	85.43%
SVM	76.60%	87.47%
Logistic Regression	74.94%	79.44%

6 Conclusion

In summary, this study has successfully demonstrated that the incorporation of emotion and sentiment analysis can significantly enhance book genre recommendations. With the use of a dataset of Tweet emotions combined with Amazon book reviews in a framework of collaborative filtering, we created a more advanced recommendation system. A key aspect of our methodology was addressing class imbalance; the application of the ADASYN technique, combined with Stratified K-Fold cross-validation, was highly effective, improving the performance of the SVM model from 76.60% on the imbalanced dataset to a stellar 87.47%. In contrast, the ENN technique failed to yield acceptable results, highlighting that not all balancing techniques are universally effective and requiring additional inquiry into its insufficiency for this specific text data. Additionally, the hyperparameter fine-tuning process was instrumental in achieving the models' full potential, with the SVM observing the most notable boost in performance, thus rendering it the supreme classifier for this task. While this research provides a firm foundation, future studies should address other algorithms like neural networks and other ensemble approaches. This study serves as an essential precursor, confirming the value of sentiment-based features and vigorous preprocessing, and so the way for more emotionally intelligent and accurate recommendation systems.

References

1. B. Pang and L. Lee, "Opinion mining and sentiment analysis," *Foundations and Trends in Information Retrieval*, vol. 2, no. 1–2, pp. 1–135, 2008.
2. B. Liu, *Sentiment Analysis and Opinion Mining*, Synthesis Lectures on Human Language Technologies. Morgan & Claypool, 2012.
3. B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment classification using machine learning techniques," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, pp. 79–86, 2002.
4. Alantari, Huwail J., et al. "An empirical comparison of machine learning methods for text-based sentiment analysis of online consumer reviews." *International Journal of Research in Marketing*, vol. 39, no. 1, 2022, pp. 1-19.
5. Chiranjeevi, Phaneendra, and A. Rajaram. "A lightweight deep learning model-based recommender system by sentiment analysis." *Journal of Intelligent & Fuzzy Systems*, Preprint, 2023, pp. 1-14.
6. Ghosh, Sourojit, et al. "Do we like this, or do we like this?": Reflections on a Human-Centered Machine Learning Approach to Sentiment Analysis." In *International Conference on Human-Computer Interaction*, pp. 63-82. Springer, 2023.
7. Devaki, P., Lokesh Kumar, and Shivali Muthukumar. "Sentiment Analysis and Recommendation of Book Reviews." In *2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)*, pp. 1-6. IEEE, 2022.
8. Santhosh, Jayasankar, et al. "Toward an Interactive Reading Experience: Deep Learning Insights and Visual Narratives of Engagement and Emotion." *IEEE Access* (2024).
9. He, Huaqian, et al. "Exploring E-Commerce Product Experience Based on Fusion Sentiment Analysis Method." *IEEE Access*, vol. 10, 2022, pp. 110248-110260.
10. Alatrash, Rawaa, and Rojalina Priyadarshini. "Fine-Grained Sentiment-Enhanced Collaborative Filtering-Based Hybrid Recommender System." *Journal of Web Engineering*, vol. 22, no. 7, 2023, pp. 983-1035.
11. Fasha, Ezqil Fasha Bin Kamal, Pantea Keikhosrokiani, and Moussa Pourya Asl. "Opinion Mining Using Sentiment Analysis: A Case Study of Readers' Response on Long Litt Woon's The Way Through the Woods in Goodreads." In *International Conference of Reliable Information and Communication Technology*, pp. 231–242. Cham: Springer International Publishing, 2021.
12. Wayesa, Fikadu, Mesfin Leranso, Girma Asefa, and Abduljebar Kedir. "Pattern-based hybrid book recommendation system using semantic relationships." *Scientific Reports* 13, no. 1 (2023): 3693.
13. Lin, Chih-Hsueh, and Ulin Nuha. "Sentiment analysis of Indonesian datasets based on a hybrid deep-learning strategy." *Journal of Big Data* 10, no. 1 (2023): 88.
14. Addanki, Mounika, S. Saraswathi, DILLI BABU SLAVAKKAM, Ramesh Babu Challagundla, and Rajendra Pamula. "Integrating Sentiment Analysis in Book Recommender System by using Rating Prediction and DBSCAN Algorithm with Hybrid Filtering Technique." (2023).
15. Karabila, Ikram, Nossayba Darraz, Anas El-Ansari, Nabil Alami, and Mostafa El Mallahi. "Enhancing collaborative filtering-based recommender system using sentiment analysis." *Future Internet* 15, no. 7 (2023): 235.
16. Dogra, V., Singh, A., Verma, S., Kavita, Jhanjhi, N. Z., & Talib, M. N. (2021). Analyzing distilbert for sentiment classification of banking financial news. In *Intelligent Computing and Innovation on Data Science: Proceedings of ICTIDS 2021* (pp. 501-510). Singapore: Springer Nature Singapore.
17. Nguyen, D. Q., Vu, T., & Nguyen, A. T. (2020). BERTweet: A pre-trained language model for English Tweets. arXiv preprint arXiv:2005.10200.
18. Fan, Y., Li, P., Kong, F., & Zhu, Q. (2025). Enhancing Multiparty Dialog Discourse Parsing With Dynamic Task-Adaptive Graph Transformer and Difficulty-Aware Task Scheduling. *IEEE Transactions on Neural Networks and Learning Systems*.
19. Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., ... & Stoyanov, V. (2019). Unsupervised cross-lingual representation learning at scale. arXiv preprint arXiv:1911.02116.
20. Belogianni, Paraskevi. "Sentiment analysis applied on a book recommendation system." (2021).
21. Singh, Jaspreet, Gurvinder Singh, and Rajinder Singh. "Optimization of sentiment analysis using machine learning classifiers." *Human-centric Computing and Information Sciences* 7 (2017): 1–12.
22. Zhou, Jie, Junfeng Tian, Rui Wang, Yuanbin Wu, Wenming Xiao, and Liang He. "Sentix: A sentiment-aware pre-trained model for cross-domain sentiment analysis." In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 568–579. 2020.
23. Walsh, Melanie, and Maria Antoniak. "The Goodreads 'Classics': A computational study of readers, Amazon, and crowdsourced amateur criticism." *Journal of Cultural Analytics* 6, no. 2 (2021).
24. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Draft, 2023.
25. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. O'Reilly Media, 2009.
26. El-Masri, Mazen, Nabeela Altrabsheh, and Hanady Mansour. "Successes and challenges of Arabic sentiment analysis research: a literature review." *Social Network Analysis and Mining* 7 (2017): 1–22.
27. Mehanna, Yassin S., and M. Mahmuddin. "The effect of pre-processing techniques on the accuracy of sentiment analysis using bag-of-concepts text representation." *SN Computer Science* 2, no. 4 (2021): 237.

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28. B. Han and T. Baldwin, "Lexical normalisation of short text messages: Makn sens a twitter," in Proc. 49th Annu. Meeting of the Association for Computational Linguistics (ACL), pp. 368–378, 2011.
 29. R. Sproat, A. Black, S. Chen, S. Kumar, M. Ostendorf, and C. Richards, "Normalization of non-standard words," *Computer Speech & Language*, vol. 15, no. 3, pp. 287–333, 2001. Saranya, S., and G. Usha. "A Machine Learning-Based Technique with IntelligentWordNet Lemmatize for Twitter Sentiment Analysis." *Intelligent Automation & Soft Computing* 36, no. 1 (2023).
 30. Hasan, Mehedi, Shenjuti Paul Puja, Md Hasan Imam Bijoy, Abdus Sattar, and Md Mahbubur Rahman. "Multiple Bangla Sentence Classification using Machine Learning and Deep Learning Algorithms." In 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT), pp. 1-6. IEEE, 2022.
 31. C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.
 32. Chiny, Mohamed, Marouane Chihab, Omar Bencharef, and Younes Chihab. "LSTM, VADER and TF-IDF based hybrid sentiment analysis model." *International Journal of Advanced Computer Science and Applications* 12, no. 7 (2021).
 33. Lavanya, P. M., and E. Sasikala. "Enhanced performance of drug review classification from social networks by improved ADASYN training and Natural Language Processing techniques." In *Computational Intelligence Methods for Sentiment Analysis in Natural Language Processing Applications*, pp. 111–127. Morgan Kaufmann, 2024.
 34. Cunningham, Padraig, and Sarah Jane Delany. "k-Nearest neighbour classifiers: (with Python examples)." arXiv preprint arXiv:2004.04523 (2020).
 35. D. L. Wilson, "Asymptotic properties of nearest neighbor rules using edited data," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-2, no. 3, pp. 408–421, 1972.
 36. L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
 37. C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
 38. D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Wiley, 2013.
 39. R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in Proc. 14th Int. Joint Conf. Artificial Intelligence (IJCAI), vol. 2, pp. 1137–1143, 1995.
 40. D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," *Journal of Machine Learning Technologies*, vol. 2, no. 1, pp. 37–63, 2011
 41. T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.
 42. K. Kowsari, K. J. Jafari Meimandi, M. Heidarysafa, S. Mendu, L. Barnes, and D. Brown, "Text classification algorithms: A survey," *Information*, vol. 10, no. 4, pp. 150, 2019.
 43. P. Lops, M. de Gemmis, and G. Semeraro, "Content-based recommender systems: State of the art and trends," in *Recommender Systems Handbook*, F. Ricci, L. Rokach, B. Shapira, and P. B. Kantor, Eds. Springer, 2011, pp. 73–105.
 44. D. Parra and S. Sahebi, "Recommender systems for literature review and citation analysis," *ACM SIGIR Forum*, vol. 47, no. 2, pp. 38–55, 2013.
 45. Baker, W., Colditz, J. B., Dobbs, P. D., Mai, H., Visweswaran, S., Zhan, J., & Primack, B. A. (2022). Classification of Twitter vaping discourse using BERTweet: comparative deep learning study. *JMIR Medical Informatics*, 10(7), e33678..
 46. Li, B., He, Y., & Xu, W. (2021). Cross-lingual named entity recognition using parallel corpus: A new approach using xlm-roberta alignment. arXiv preprint arXiv:2101.11112.
 47. Pang, B., Lee, L., & Vaithyanathan, S. (2002). Thumbs up? Sentiment classification using machine learning techniques. arXiv preprint cs/0205070.
 48. Go, A., Bhayani, R., & Huang, L. (2009). Twitter sentiment classification using distant supervision. CS224N project report, Stanford, 1(12), 2009.
 49. Liu, B., Hu, M., & Cheng, J. (2005, May). Opinion observer: analyzing and comparing opinions on the web. In *Proceedings of the 14th international conference on World Wide Web* (pp. 342-351).
 50. Turney, P. D. (2002). Thumbs up or thumbs down? Semantic orientation applied to unsupervised classification of reviews. arXiv preprint cs/0212032.